

Digital Design (SET 111)

L11

Latches, Flip-Flops, and Timers

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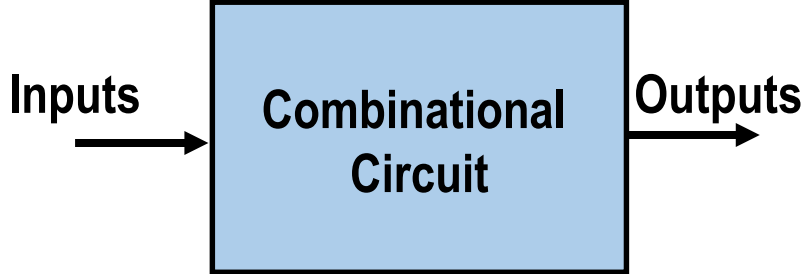
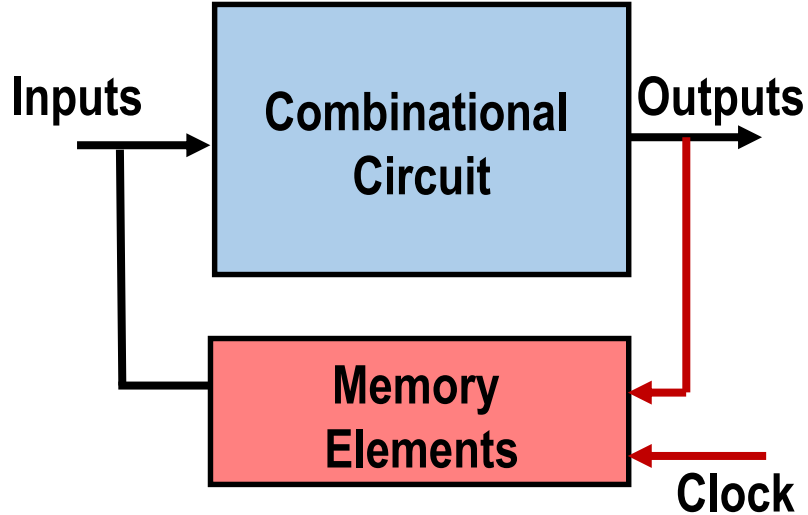
- ☐ **Types of digital logic circuits**
- ☐ **Types of Storage Elements**
- ☐ **SR-Latch**
- ☐ **Gated SR-Latch**
- ☐ **D-Latch**
- ☐ **D-Flip-Flop**
- ☐ **JK-Flip-Flop**
- ☐ **Flip-Flop Applications**
- ☐ **555 Timer**



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- ☐ **JK-Flip-Flop**
- ☐ **Flip-Flop Applications**
- ☐ **555 Timer**

Types of Digital Logic Circuits



Combinational Circuit	Sequential Circuit
 Output depends on the present state of input only.	 Output depends on the present inputs and the past history of inputs.
Does not contain memory elements.	Contains memory elements.
Does not use the Feedback path	Uses Feedback path.
Does not require a clock signal.	Most sequential circuits use a clock signal.
Ex. Adder, Subtractor, MUX, Decoder, Encoder..etc	Ex. Flip Flops, Registers, Counters, etc

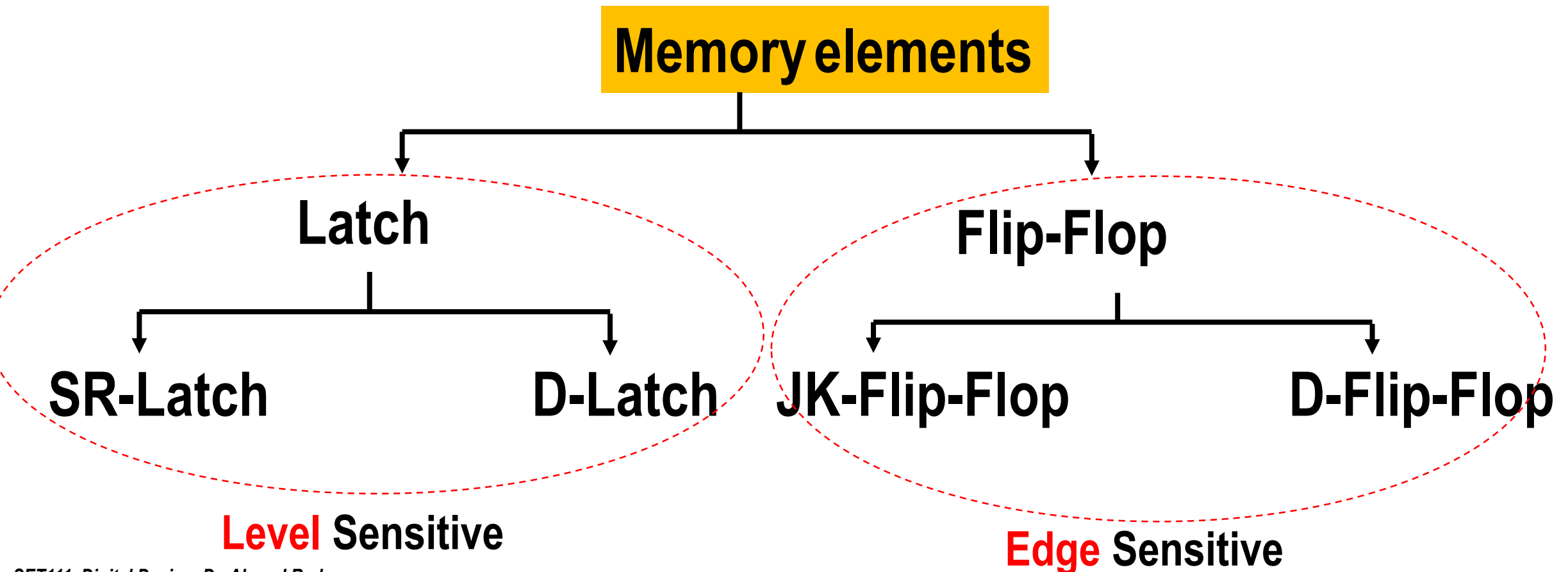


- ☐ Types of digital logic circuits
- ☐ **Types of Storage Elements**
- ☐ SR-Latch
- ☐ Gated SR-Latch
- ☐ D-Latch
- ☐ D-Flip-Flop
- ☐ JK-Flip-Flop
- ☐ Flip-Flop Applications
- ☐ 555 Timer

Types of Storage Elements (1/3)



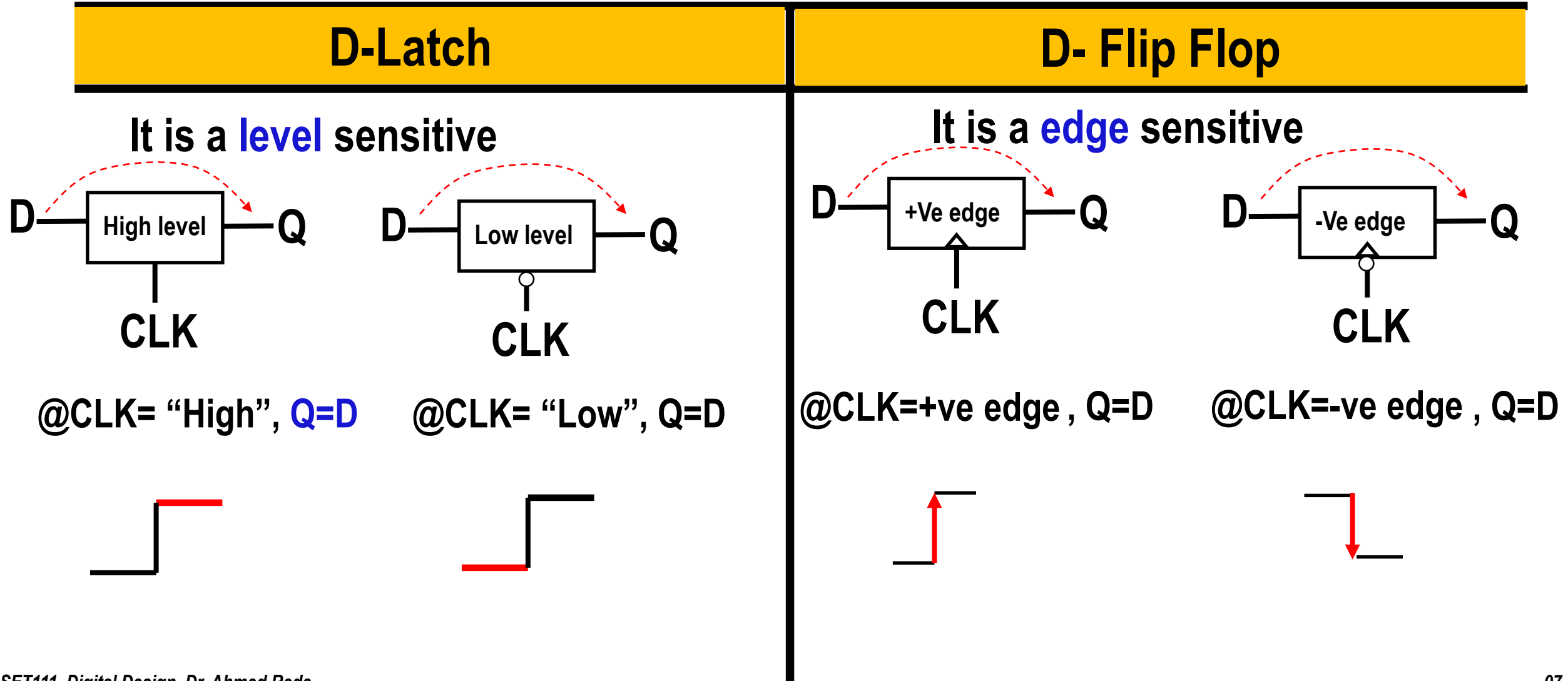
□ **Storage/Memory** elements are needed in most digital logic circuits to hold (**remember**) logic values.



Types of Storage Elements (2/3)



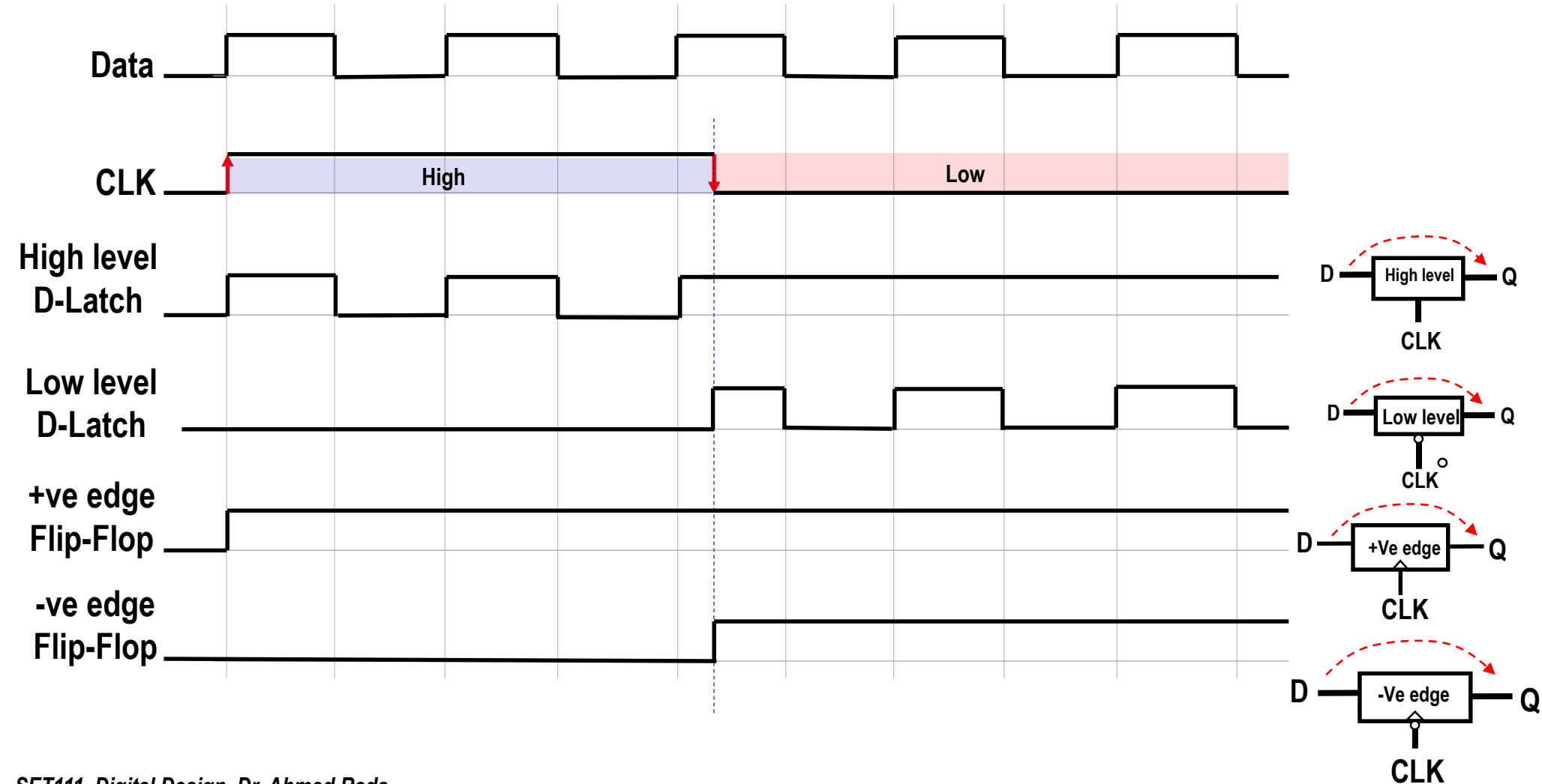
- ❑ To understand the difference between the latch and flip-flop.



Types of Storage Elements (3/3)



□ Timing diagram for D-Latch and D-Flip-flop



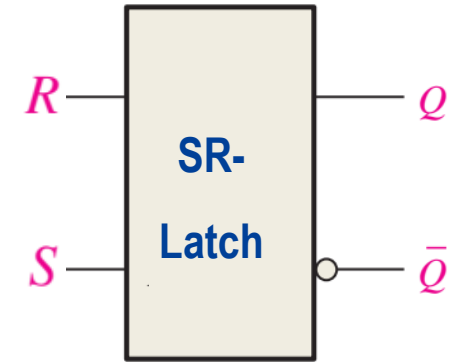


- ☐ Types of digital logic circuits
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- ☐ 555 Timer

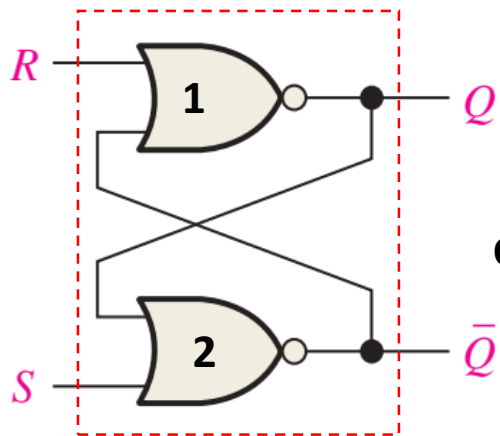
SR-Latch (1/11)



□ Active-HIGH input S-R latch



Block Diagram



Two cross-coupled NOR gates

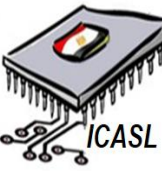
Internal Circuit

Truth Table

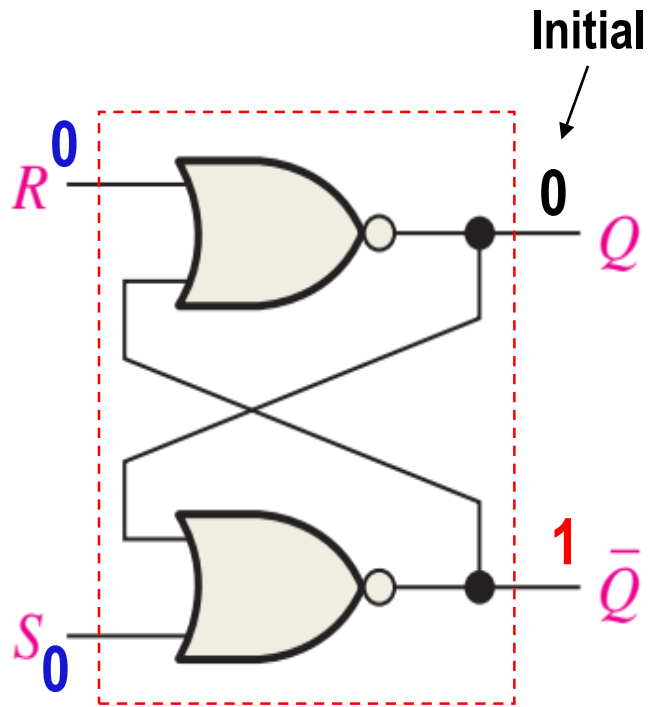
Inputs		Output		Comments
S	R	Q	$\bar{Q}$	
0	0	Q_0	$\bar{Q}_0$	No change, Latch remains in present state
0	1	0	1	RESET
1	0	1	0	SET
1	1	0	0	Invalid Condition

Practically, setting both inputs to 1 is **forbidden!**

SR-Latch (2/11)



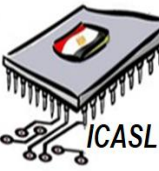
❑ Operation of Active-HIGH input S-R latch



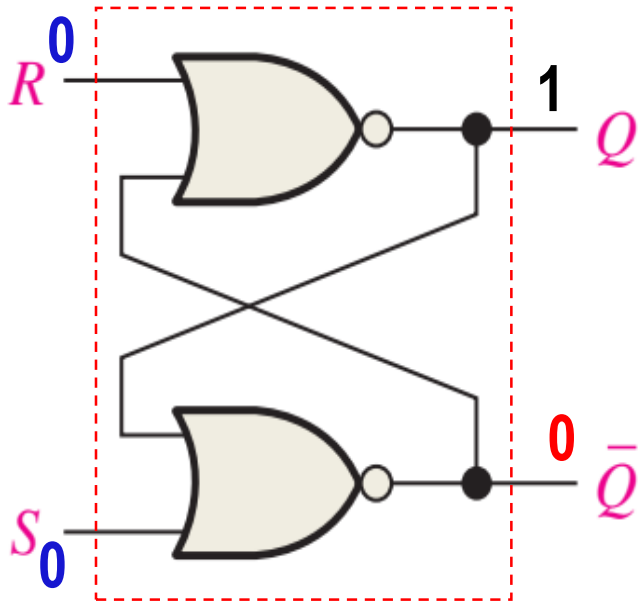
Truth Table

Inputs		Initial Output	Output		Comments
S	R	Q_0	Q	$\overline{Q}$	
0	0	0	0	1	
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

SR-Latch (3/11)



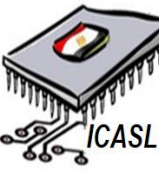
□ Operation of Active-HIGH input S-R latch



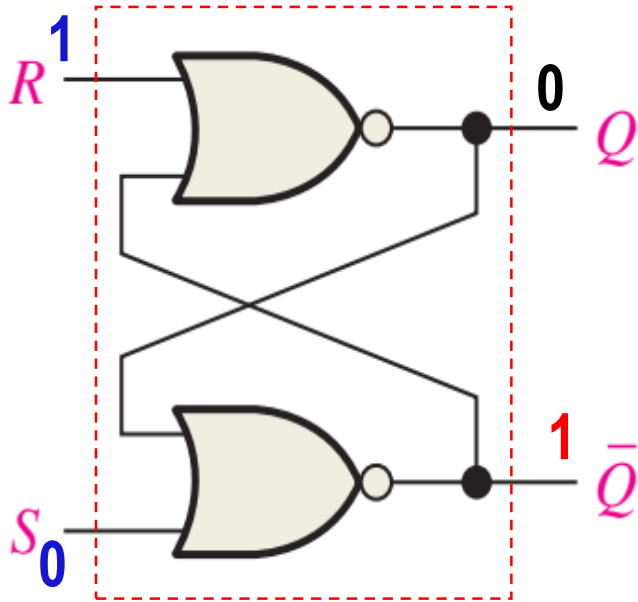
Truth Table

Inputs		Initial Output	Output		Comments
S	R	Q_0	Q	$\overline{Q}$	
0	0	0	0	1	No change, Latch remains in present state
0	0	1	1	0	
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

SR-Latch (4/11)



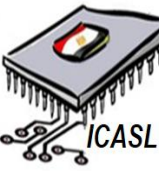
❑ Operation of Active-HIGH input S-R latch



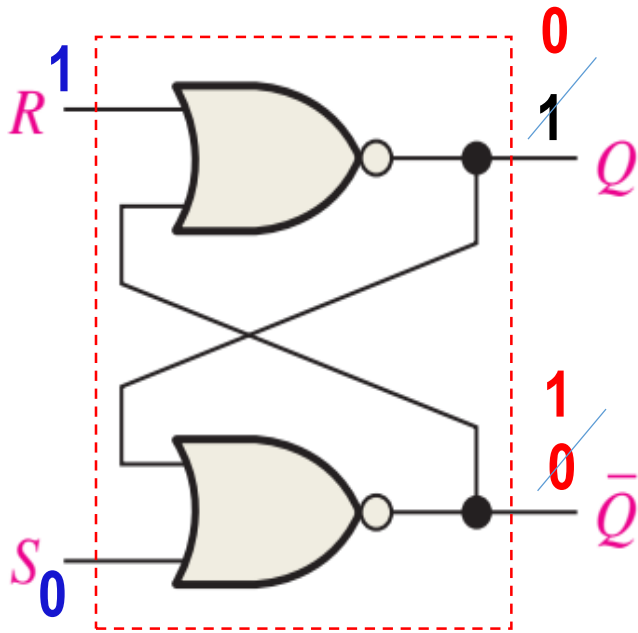
Truth Table

Inputs		Initial Output	Output		Comments
S	R	Q_0	Q	$\overline{Q}$	
0	0	0	0	1	No change, Latch remains in present state
0	0	1	1	0	
0	1	0	0	1	
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

SR-Latch (5/11)



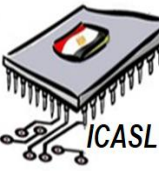
❑ Operation of Active-HIGH input S-R latch



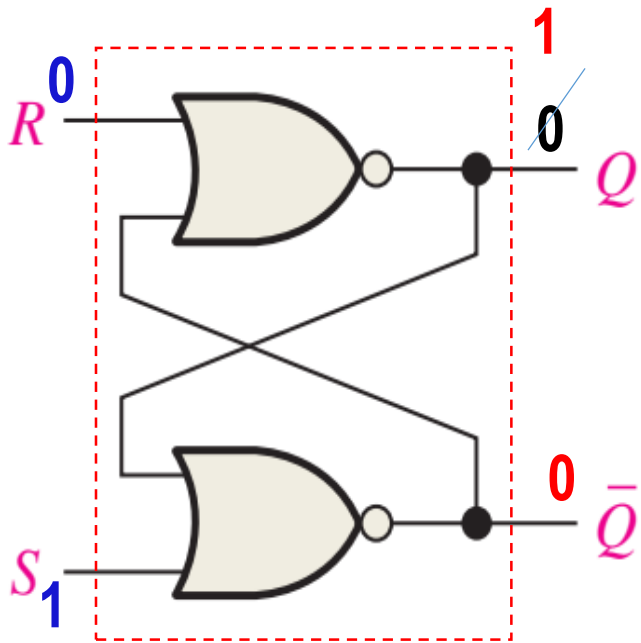
Truth Table

Inputs		Initial Output	Output		Comments
S	R	Q_0	Q	$\overline{Q}$	
0	0	0	0	1	No change, Latch remains in present state
0	0	1	1	0	
0	1	0	0	1	RESET
0	1	1	0	1	
1	0	0			
1	0	1			
1	1	0			
1	1	1			

SR-Latch (6/11)



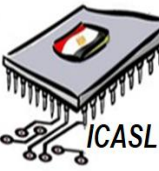
❑ Operation of Active-HIGH input S-R latch



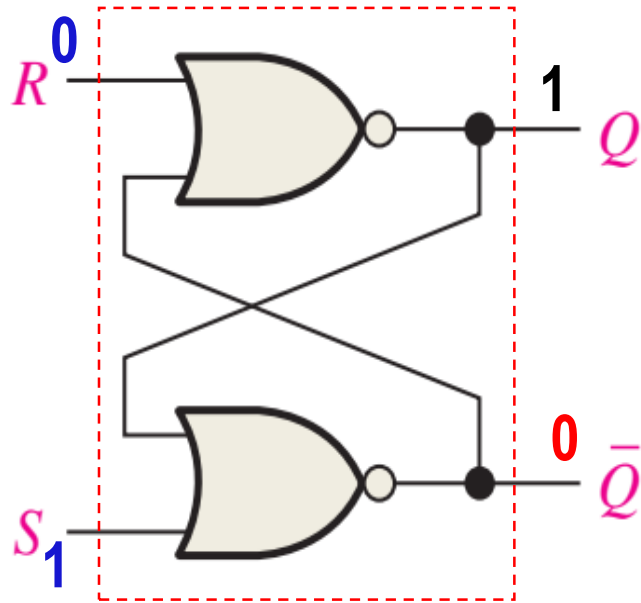
Truth Table

Inputs		Initial Output	Output		Comments
S	R	Q_0	Q	$\overline{Q}$	
0	0	0	0	1	No change, Latch remains in present state
0	0	1	1	0	
0	1	0	0	1	RESET
0	1	1	0	1	
1	0	0	1	0	
1	0	1			
1	1	0			
1	1	1			

SR-Latch (7/11)



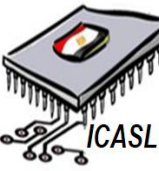
□ Operation of Active-HIGH input S-R latch



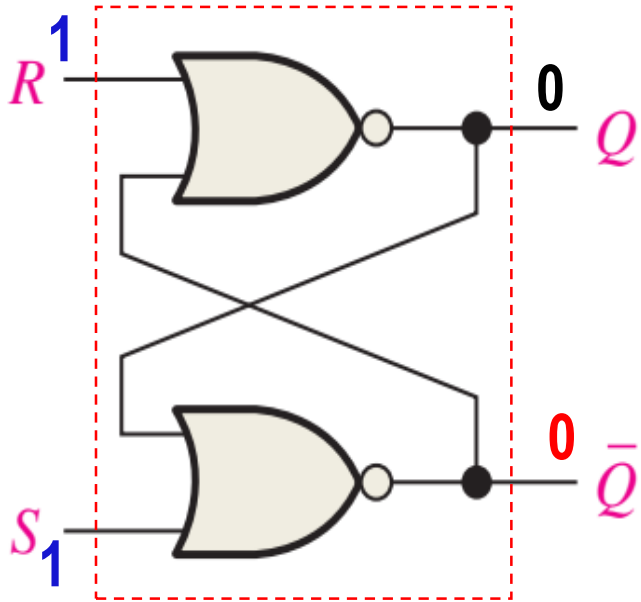
Truth Table

Inputs		Initial Output	Output		Comments
S	R	Q_0	Q	$\overline{Q}$	
0	0	0	0	1	No change, Latch remains in present state
0	0	1	1	0	
0	1	0	0	1	RESET
0	1	1	0	1	
1	0	0	1	0	SET
1	0	1	1	0	
1	1	0			
1	1	1			

SR-Latch (8/11)



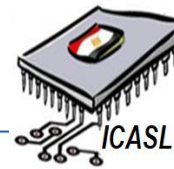
□ Operation of Active-HIGH input S-R latch



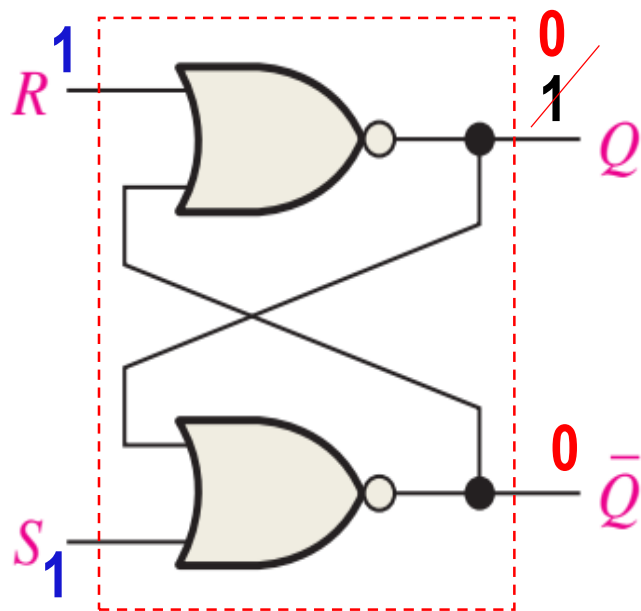
Truth Table

Inputs		Initial Output	Output		Comments
S	R	Q_0	Q	$\overline{Q}$	
0	0	0	0	1	No change, Latch remains in present state
0	0	1	1	0	
0	1	0	0	1	RESET
0	1	1	0	1	
1	0	0	1	0	SET
1	0	1	1	0	
1	1	0	0	0	
1	1	1			

SR-Latch (9/11)



❑ Operation of Active-HIGH input S-R latch



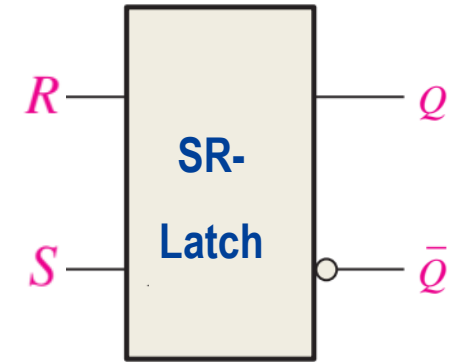
Truth Table

Inputs		Initial Output	Output		Comments
S	R	Q_0	Q	$\overline{Q}$	
0	0	0	0	1	No change, Latch remains in present state
0	0	1	1	0	
0	1	0	0	1	RESET
0	1	1	0	1	
1	0	0	1	0	SET
1	0	1	1	0	
1	1	0	0	0	Invalid Condition
1	1	1	0	0	

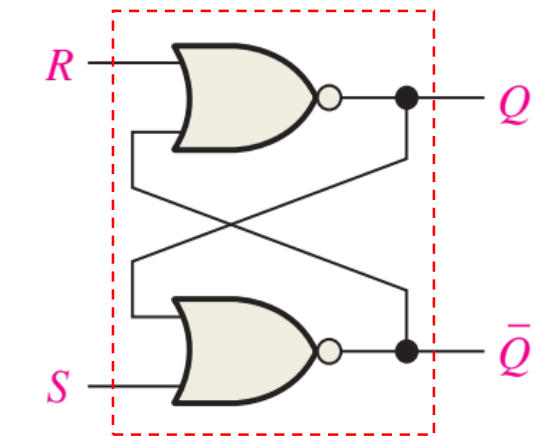
SR-Latch (10/11)



❑ Operation of Active-HIGH input S-R latch



Block Diagram



Internal Circuit

Truth Table

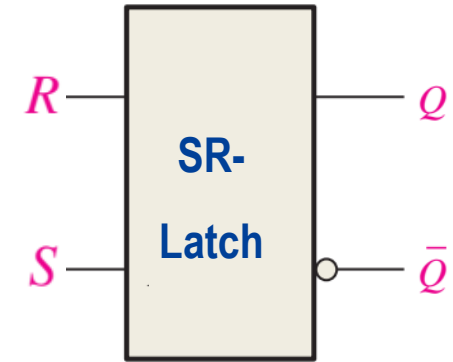
Inputs		Initial Output	Output		Comments
S	R	Q_0	Q	$\overline{Q}$	
0	0	0	0	1	No change, Latch remains in present state
0	0	1	1	0	
0	1	0	0	1	RESET
0	1	1	0	1	
1	0	0	1	0	SET
1	0	1	1	0	
1	1	0	0	0	Invalid Condition
1	1	1	0	0	

Practically, setting both inputs to 1 is forbidden!

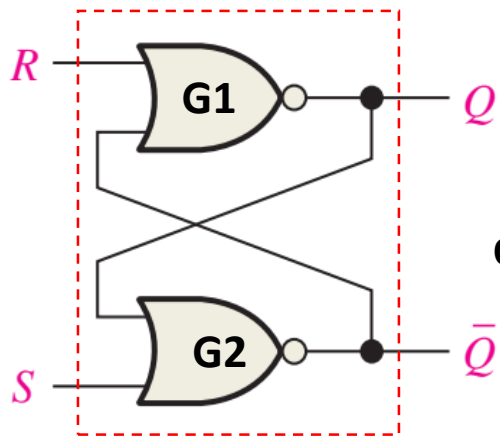
SR-Latch (11/11)



□ Active-HIGH input S-R latch → **Summary**



Block Diagram



Internal Circuit

Truth Table

Inputs		Output		Comments
S	R	Q	$\bar{Q}$	
0	0	Q_0	$\bar{Q}_0$	No change, Latch remains in present state
0	1	0	1	RESET
1	0	1	0	SET
1	1	0	0	Invalid Condition

Practically, setting both inputs to 1 is **forbidden!**

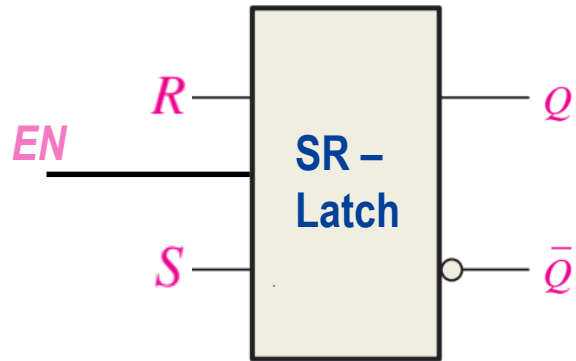


- ☐ Types of digital logic circuits
- ☐ Types of Storage Elements
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- ☒ Gated SR-Latch
- ☐ D-Latch
- ☐ D-Flip-Flop
- ☐ JK-Flip-Flop
- ☐ Flip-Flop Applications
- ☐ 555 Timer

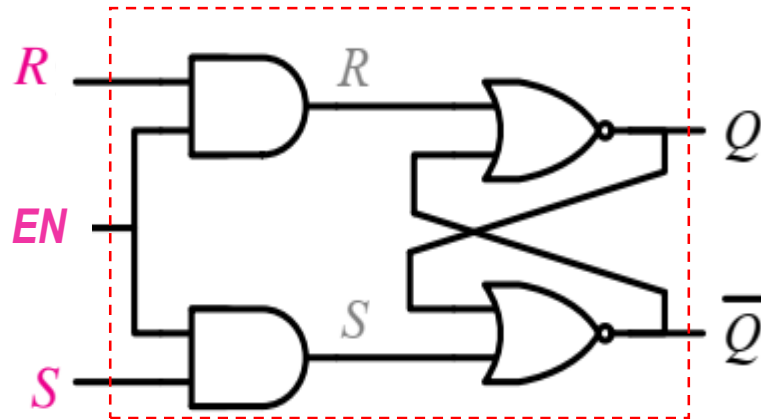
Gated SR-Latch



❑ A gated latch requires an **Enable** input



Block Diagram



Internal Circuit

Truth Table

Inputs			Output		Comments
EN	S	R	Q	$\overline{Q}$	
1	0	0	Q_0	$\overline{Q_0}$	No change, Latch remains in present state
1	0	1	0	1	RESET
1	1	0	1	0	SET
1	1	1	0	0	Invalid Condition
0	X	X	Q_0	$\overline{Q_0}$	No change, Latch remains in present state



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D-Latch (1/3)



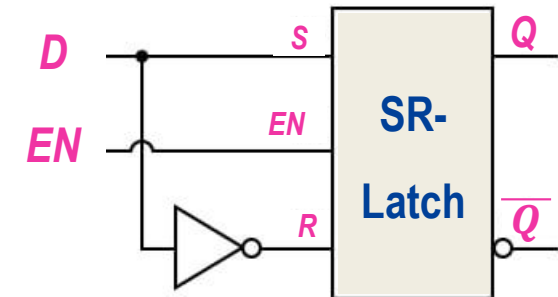
□ D-latch has **only one** input in addition to EN.

□ This input is called the D (data) input.

□ When the D input is **HIGH** and the EN input is **HIGH**, the latch will set ($Q=D$).

□ When the D input is **LOW** and EN is **HIGH**, the latch will reset ($Q=0$).

The output Q follows the input D when EN is HIGH.



Block Diagram

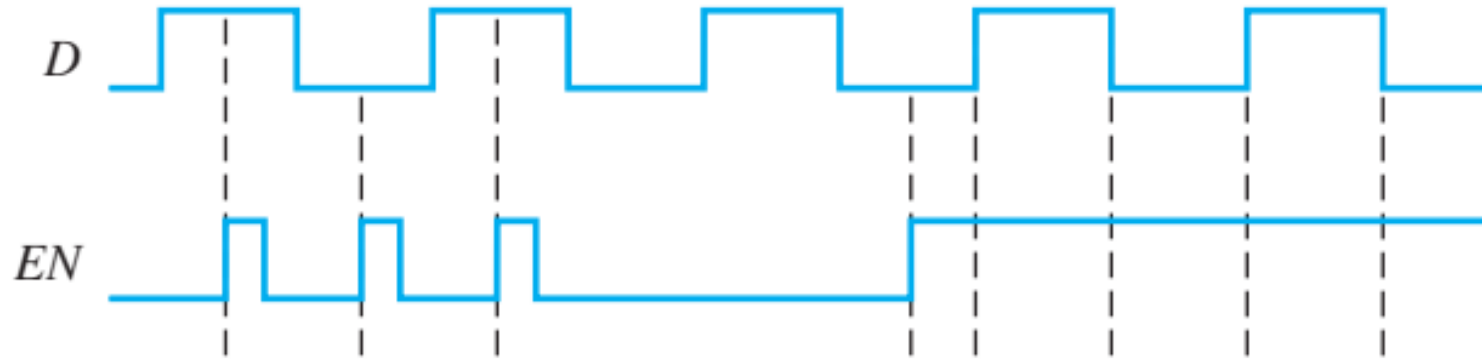
Truth Table

Inputs		Output		Comments
EN	D	Q	$\overline{Q}$	
1	0	0	1	Q=D
1	1	1	0	
0	X	Q_0	Q_0	No change

D-Latch (2/3)



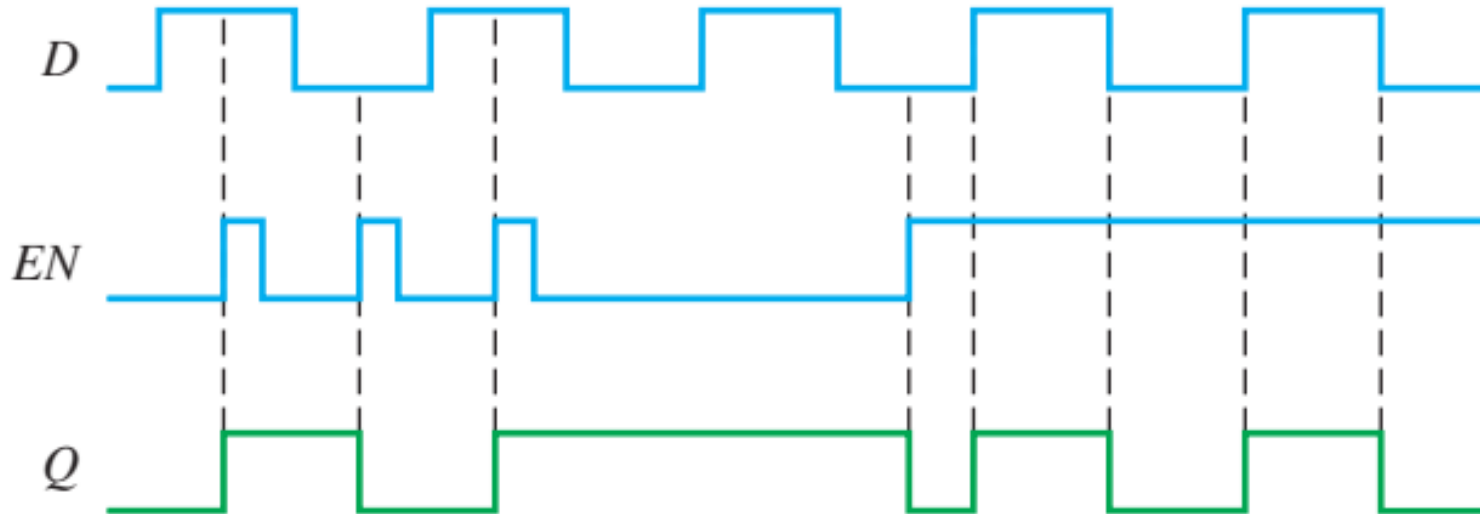
□ Ex.1 Determine the Q output waveform if the inputs shown in Figure are applied to a gated D latch, which is initially RESET



D-Latch (3/3)



- Ex.1 Determine the Q output waveform if the inputs shown in Figure are applied to a gated D latch, which is initially RESET



Solution

- *As Long as $EN=1$ the Latch output asynchronously follows input Data D*
- *When EN is LOW, the state of the latch is not affected by the D input*



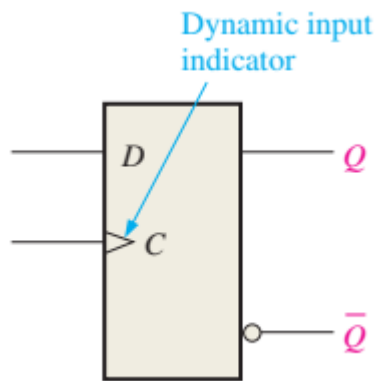
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Flip-Flop

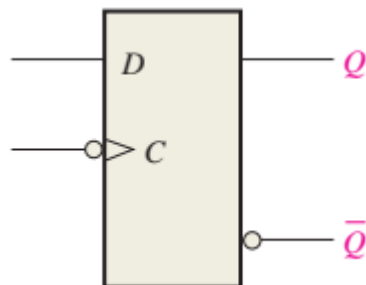


❑ The edge-sensitive flip-flop changes state either at the **positive** or at the **negative** edge of the clock pulse.

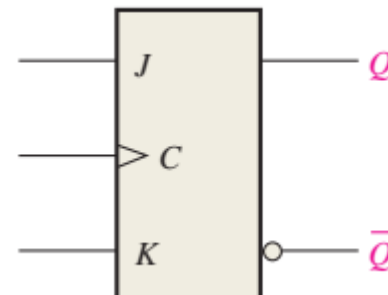
❑ **Two types** of edge-sensitive flip-flops are D-Flip-flop and J-K- Flip-Flop



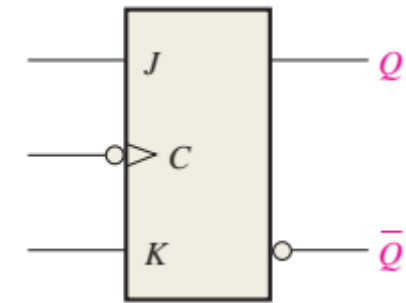
D-Flip-Flop
Positive edge



D-Flip-Flop
Negative edge



JK -Flip-Flop
Positive edge



JK -Flip-Flop
Negative edge

D-Flip-Flop (1/3)

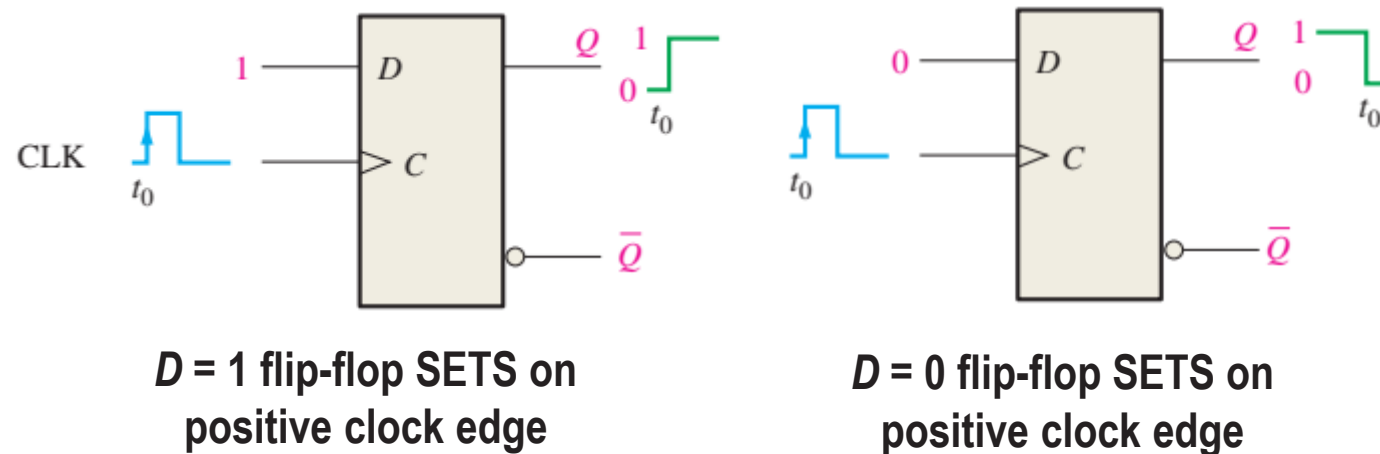


❑ The D input of the D-flip-flop is a **synchronous input** because data on the input are transferred to the flip-flop's output **only** on the triggering edge of the clock.

❑ When D is **HIGH**, the Q output goes **HIGH** on the triggering edge of the clock pulse, and the flip-flop is **SET**.

❑ When D is **LOW**, the Q output goes **LOW** on the triggering edge of the clock pulse, and the flip-flop is **RESET**.

The output Q follows the input D @ triggering edge of the clock



Truth Table of +ve edge-triggered D-Flip Flop

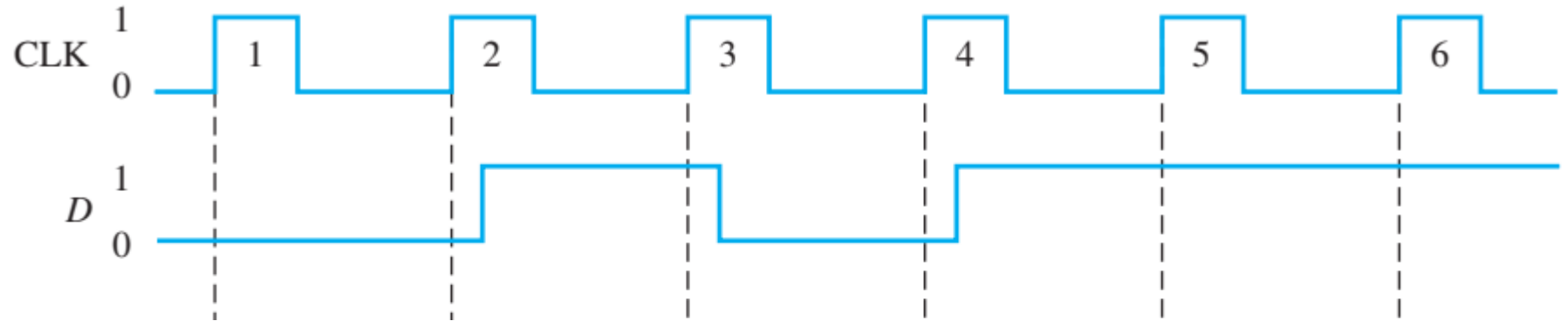
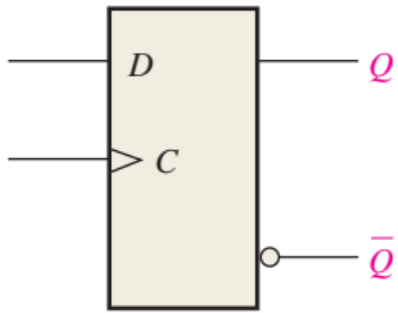
Inputs		Outputs		Comments
D	CLK	Q	$\bar{Q}$	
0	↑	0	1	RESET
1	↑	1	0	SET

↑: Clock transition LOW to HIGH

D-Flip-Flop (2/3)



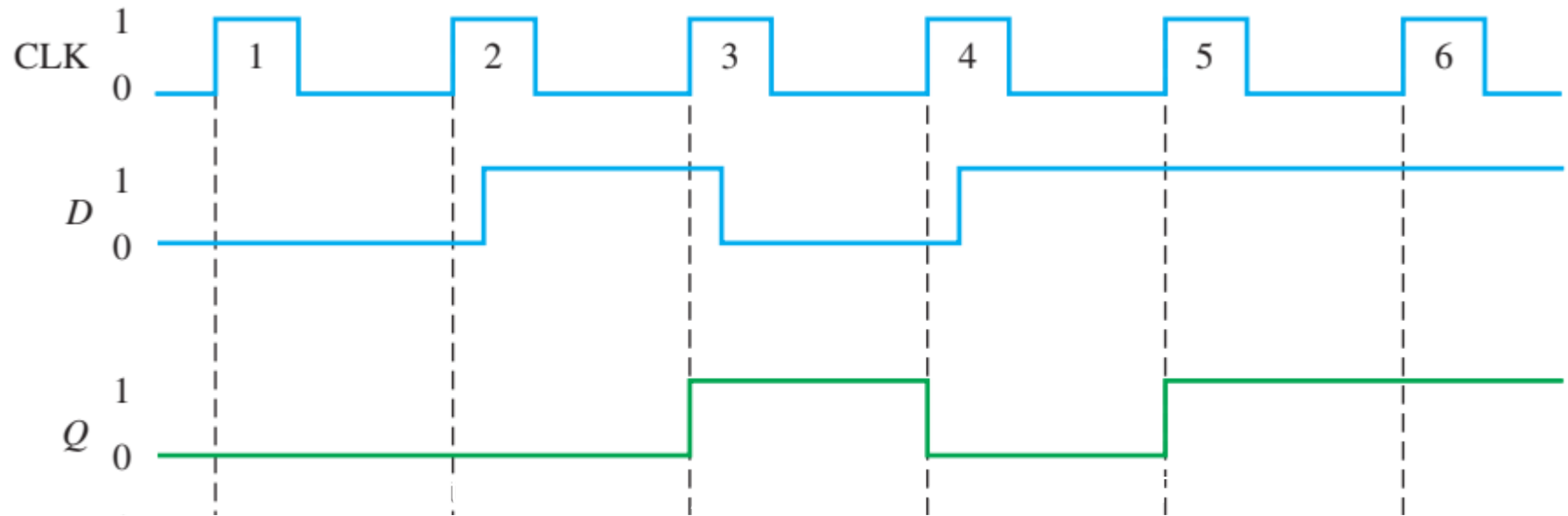
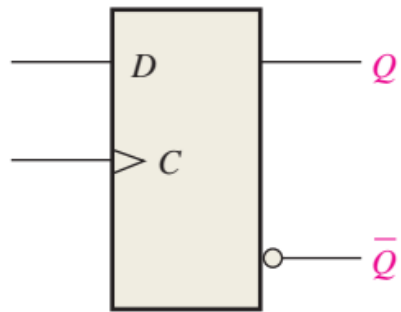
□ Determine the Q and $\overline{Q}$ output waveforms of the flip-flop in Figure for the D and CLK inputs in Figure. Assume that the positive edge-triggered flip-flop is initially RESET.



D-Flip-Flop (2/3)



□ Determine the Q and $\overline{Q}$ output waveforms of the flip-flop in Figure for the D and CLK inputs in Figure. Assume that the positive edge-triggered flip-flop is initially RESET.



Solution

1. At clock pulse 1, D is LOW, so Q remains LOW (RESET).
2. At clock pulse 2, D is LOW, so Q remains LOW (RESET).
3. At clock pulse 3, D is HIGH, so Q goes HIGH (SET).
4. At clock pulse 4, D is LOW, so Q goes LOW (RESET).
5. At clock pulse 5, D is HIGH, so Q goes HIGH (SET).
6. At clock pulse 6, D is HIGH, so Q remains HIGH (SET).

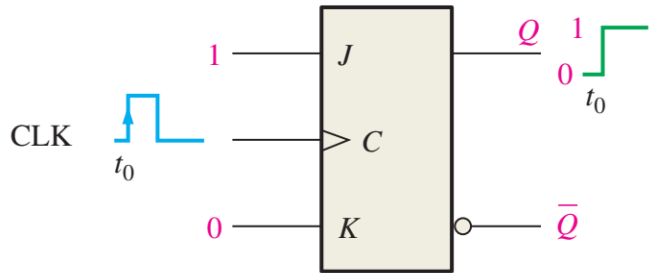


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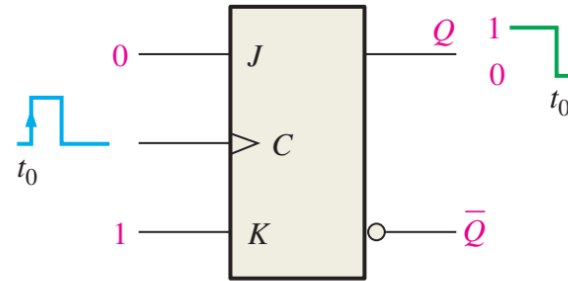
JK-Flip-Flop



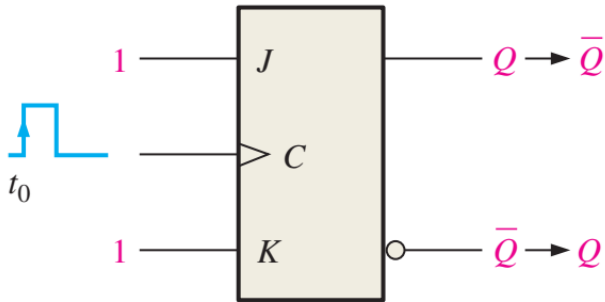
❑ The J and K inputs of the J-K flip-flop are synchronous inputs because data on these inputs are transferred to the flip-flop's output only on the triggering edge of the clock pulse.



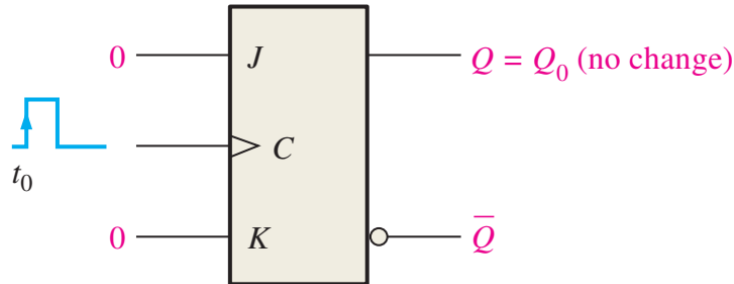
(a) $J = 1, K = 0$ flip-flop SETS on positive clock edge. (If already SET, it remains SET.)



(b) $J = 0, K = 1$ flip-flop RESETS on positive clock edge. (If already RESET, it remains RESET.)



(c) $J = 1, K = 1$ flip-flop changes state (toggle).



(d) $J = 0, K = 0$ flip-flop does not change. (If SET, it remains SET; if RESET, it remains RESET.)

Truth Table of +ve edge-triggered JK -Flip Flop

Inputs			Outputs		Comments
j	k	CLK	Q	$\bar{Q}$	
0	0	↑	Q_0	$\bar{Q}_0$	No change
0	1	↑	0	1	RESET
1	0	↑	1	0	SET
1	1	↑	$\bar{Q}_0$	Q_0	Toggle

↑: Clock transition LOW to HIGH



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- ☐ 555 Timer

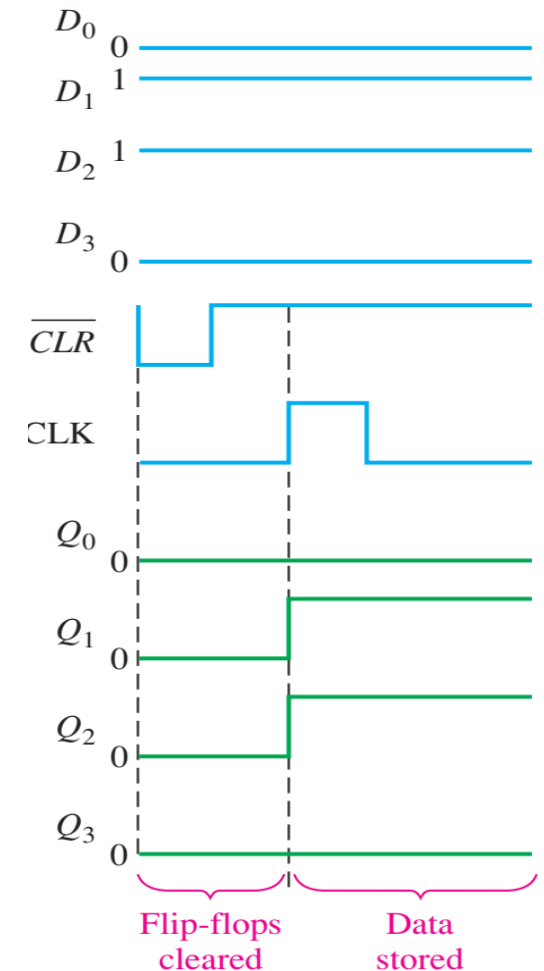
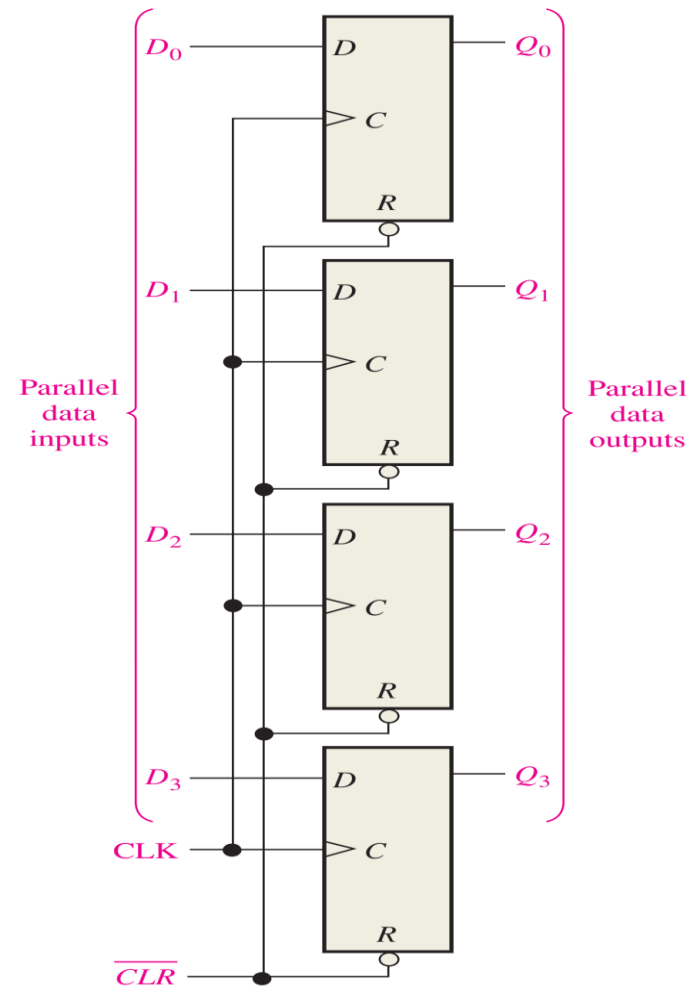
Flip-Flop Applications (1/2)



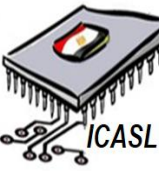
1. Parallel Data Storage

- A common requirement in digital systems is to store several bits of data from parallel lines simultaneously in a group of flip-flops

This group of 4 flip-flops is an example of a basic register used for data storage. In digital systems, data are normally stored in groups of bits (usually 8 or multiples thereof) that represent numbers, codes, or other information.

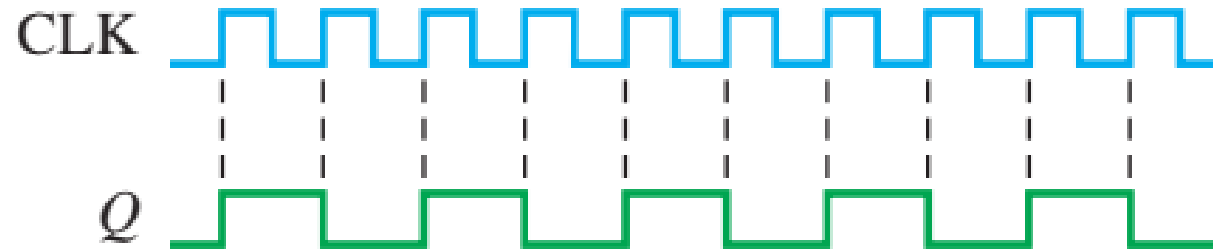
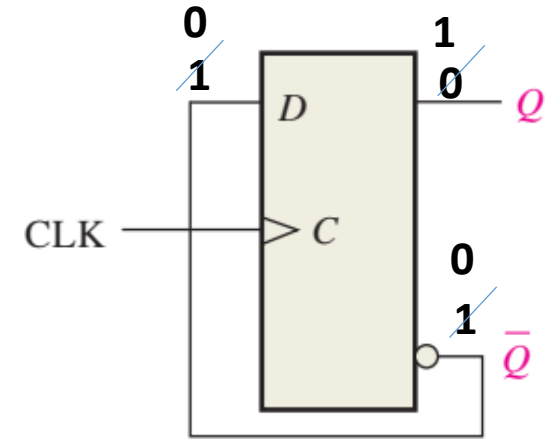


Flip-Flop Applications (2/2)

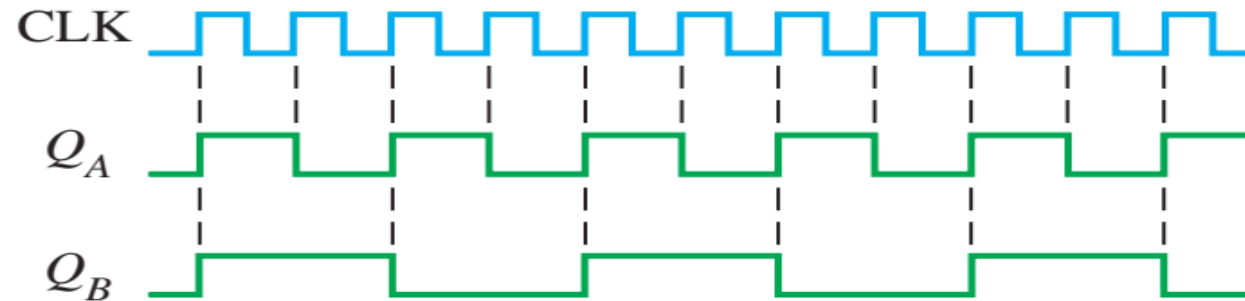
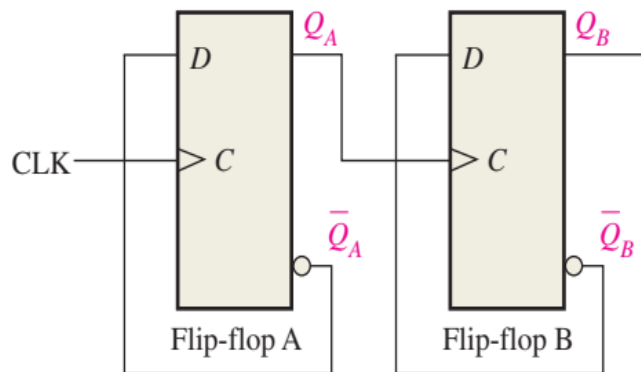


□ 2. Frequency Division

□ Another application of a flip-flop is dividing (reducing) the frequency of a periodic waveform



Q is one-half the frequency of CLK



Q_A is one-half and Q_B is one-fourth the frequency of CLK.

By connecting flip-flops in this way, a frequency division of 2^n is achieved, where n is the number of flip-flops. For example, three flip-flops divide the clock frequency by $2^3 = 8$; four flip-flops divide the clock frequency by $2^4 = 16$; and so on.

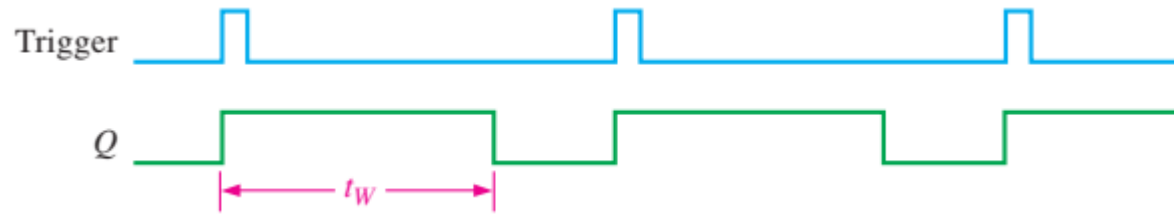
- ☐ Types of digital logic circuits
- ☐ Types of Storage Elements
- ☐ SR-Latch
- ☐ Gated SR-Latch
- ☐ D-Latch
- ☐ D-Flip-Flop
- ☐ JK-Flip-Flop
- ☐ Flip-Flop Applications
- ☐ **555 Timer**

The 555 Timer (1/3)



❑ The 555 **timer** is a versatile and widely used IC device because it can be configured in two different modes

❑ A monostable multivibrator (one-shot).



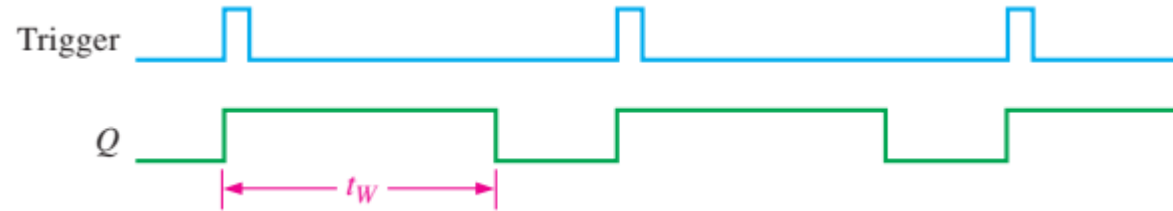
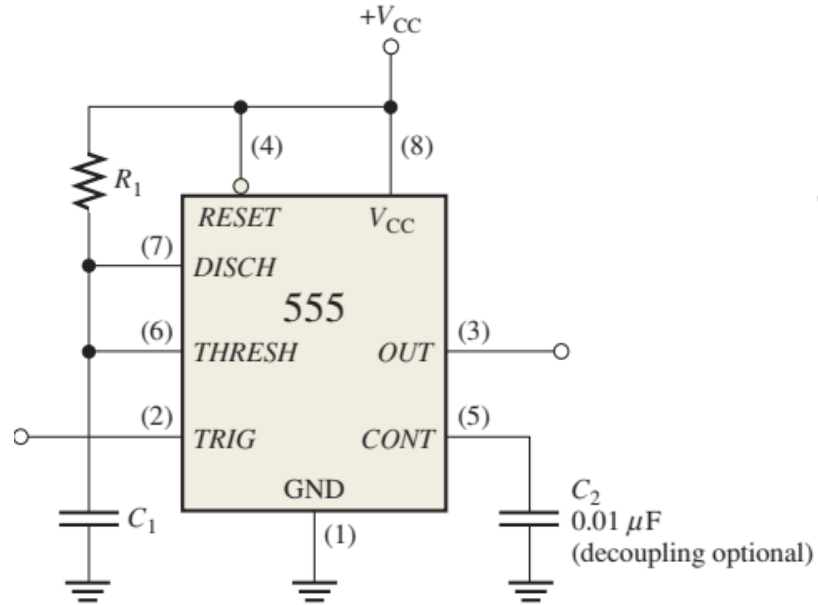
❑ An astable multivibrator (pulse oscillator).



The 555 Timer (2/3)



❑ A mono-stable multi-vibrator (one-shot).



$$t_w = 1.1R_1C_1$$

EXAMPLE

What is the output pulse width for a 555 monostable circuit with $R_1 = 2.2 \text{ k}\Omega$ and $C_1 = 0.01 \text{ }\mu\text{F}$?

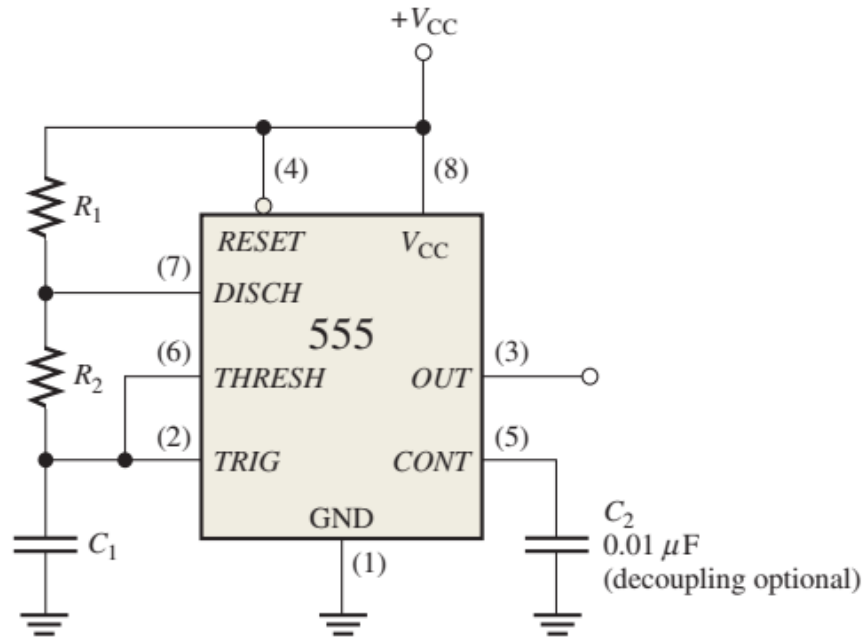
Solution

$$t_w = 1.1R_1C_1 = 1.1(2.2 \text{ k}\Omega)(0.01 \text{ }\mu\text{F}) = 24.2 \text{ }\mu\text{s}$$

The 555 Timer (3/3)



❑ An **astable** multi-vibrator.



$$f = \frac{1.44}{(R_1 + 2R_2)C_1}$$

$$\text{Duty cycle} = \left(\frac{R_1}{R_1 + R_2} \right) 100\%$$

EXAMPLE

$$f = \frac{1.44}{(R_1 + 2R_2)C_1} = \frac{1.44}{(2.2 \text{ k}\Omega + 9.4 \text{ k}\Omega)0.022 \text{ }\mu\text{F}} = 5.64 \text{ kHz}$$

$$\text{Duty cycle} = \left(\frac{R_1 + R_2}{R_1 + 2R_2} \right) 100\% = \left(\frac{2.2 \text{ k}\Omega + 4.7 \text{ k}\Omega}{2.2 \text{ k}\Omega + 9.4 \text{ k}\Omega} \right) 100\% = 59.5\%$$

Thanks!